5th Semester		
ISCED Code	**Course Code**	**Course Title**
0613	CSE-3521	Computer Architecture
Credit Hours: 3	**Contact Hours:** 3	**Type:** Core, Engineering
Prerequisite: CSE-2305 (Digital Logic Design)		
Co-requisite: NA		

Course Assessments	**CIE**: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30 Marks
	SEE: Semester End Examination		50 Marks
	Total		100 Marks

Course Rationale:

The purpose of this course is to provide an overview of the architecture and organization of a computer hardware system and how it is built. Initially necessary preliminaries required for analyzing and measuring the computer performance are taught. this course familiarizes students with CPU, memory, I/O organization, peripheral and also the contemporary design and implementation of microprocessors.

Objectives:

The objectives of this course are :

a) To introduce the student with CPU, memory, I/O organization and peripherals.

b) To introduce students to the contemporary design and implementation of microprocessors.

c) To help the student to demonstrate an understanding of micro architecture concepts and features employed by contemporary microprocessors, such as Pipelining, data forwarding, super scalar, branch predictors and also competence of basic skills in optimizing assembly programs.

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

#	CLO Description	Weightage (%)
1.	**Understand** the basic organization of computers and different instruction formats and addressing modes. Understand memory organization, including cache memory, main memory, and virtual memory. Understand various processor architectures, including RISC, CISC, and modern multi-core systems.	25%
2.	**Analyze** the performance of computer systems using metrics, various issues related to memory hierarchy, various modes of data transfer	45%

	between CPU and I/O devices, the impact of memory hierarchy on performance. Analyze techniques like interrupts, direct memory access (DMA), and I/O channeling.	
3.	**Illustrate** the working of the CPU, datapath, and control signals during instruction execution through clear diagrams and examples. Use visual aids to explain complex processes like pipelining, memory management, and I/O operations effectively.	30%

Mapping of CLO-PLO-DL-KP-EP-EA:

CLOs		**PLOs**	**DL**	**KP**	**EP**	**EA**	**Teaching Learning Strategy (TLS)**	**Assessment Strategy (AS)**
CLO1	**Understand** the basic organization of computer and different instruction formats and addressing modes. Various modes of data transfer between CPU and I/O devices.	PLO1	C2	K3, K4	EP1, EP3, EP4	-	Lecture, Class discussion, Note	Exam, Quiz, Assignment
CLO2	**Analyze** the performance of computer systems using metrics, various issues related to memory hierarchy, various modes of data transfer between CPU and I/O devices, the impact of memory hierarchy on performance. Analyze techniques like interrupts, direct memory access (DMA), and I/O channeling.	PLO2	C3	K3, K4	EP1, EP3, EP4	-	Discussion, Assignment	Assignment, Quiz, Exam

CLO3	**Illustrate** the working of the CPU, datapath, and control signals during instruction execution through clear diagrams and examples. Use visual aids to explain complex processes like pipelining, memory management, and I/O operations effectively.	PLO2	C3	K2, K4	EP1, EP3, EP4	-	Lecture, Assignment, Quiz	CT, Assignment
Note: DL: Domain/level of learning taxonomy, **KP:** Knowledge Profile, **EP:** Attribute of Complex Engineering Problems, **EA:** Attribute of Complex Engineering Activities, **Learning Domains** (C: Cognitive, A: Affective, P: Psychomotor)								

Course Content:

#	Content	Duration	CLOs
Mid-Term (30 Marks)			
1	Computer architecture Basic, History, Different terminologies of computing device, Types of Computer Architecture Understanding Program Performance, Defining Performance, Measuring Performance, CPU Performance and its factor, Evaluating performance, MIPS as a performance Measure	**2 weeks**	CLO1
2	Instruction and data access methods, Instruction Set, Stored-Program Concept, Operations of the computer Hardware, Operands of the computer Hardware (Design Principles of Computer Hardware) Representation of Instructions in the Computer, Logical Operations, Instructions for decision making, MIPS Addressing for 32-Bit Immediate and Addresses	**2 weeks**	CLO1, CLO2

3	Arithmetic and logical operations, floating point operations, ALU design, Signed and Unsigned numbers, Number Conversion and representation, Arithmetic Operations and Representation Matrix-chain multiplication and longest common subsequence problems as examples, Complexity analysis of the algorithms. Multiplication, Division and Floating point Hardware	**2 weeks**	CLO2
Final Exam: 50 Marks **Group-A (20 Marks)**			
4	The control unit (Single cycle Data path) : hardwired and micro programmed, Logic Design Convention, Clocking Methodology, Data path Basic, State Element of Data path Building a single Data path (R-Type, I-Type and J-type Instructions) Designing the main control unit (with control signals effect), Operation of the data path	**2 week**	CLO2
5	The control unit (Multiple cycle Data path): Hardwired and micro programmed, Multi-cycle implementation Basic. Necessity of multi-cycle implementation, Details of Control Signal Breaking the Instruction Execution into Clock Cycles, Exceptions	**2 weeks**	CLO2
Group-B (30 Marks)			
6	Pipelining: Overview of Pipelining, Pipelining Hazard, Pipeline Data Path (R-Type, I-Type and J-type Instructions) Pipeline Control, Exception, Exception Handling	**1 week**	CLO3
7	Memory organization: Introduction to memory, memory Hierarchy, Basic on Cache, Accessing a cache, Handling cache miss and writes, Flexible placement of cache block	**2 weeks**	CLO3
8	I/O systems, channels, interrupts, DMA, I/O Devices basic, Disk Storage and dependability, RAID, Buses and other connections between processors, Memory and I/O Devices, Interfacing I/O devices with processors.	**2 week**	CLO3

	Introduction to virtual Memory, Virtual to physical, Handling page Fault TLB, Integrating Virtual Memory, TLBs and Caches		

Marks Distribution:

Course Assessment Pattern (Theory courses):

		CIE (70 marks)		**SEE (30 marks)**
Cognitive learning	Affective learning	Attendance Marks (10)	Mid-term (30)	Final Exam (50)
Remember				
Understand			5	5
Apply			5	5
Analyze			5	5
Evaluation			10	15
Create			5	5
	Responding	10		
Total allocated marks		10	30	50

Grading Policy: As per IIUC grading policy

Learning Materials:

#	Name of Authors	Title of Book	Edition	Publisher's Name	Year	ISBN
1	John Hennesy, David Patterson	Computer Organization	**3rd Edition**	Morgan Kaufman publisher	2005	ISBN: 1-55860- 604-1
2	William Stallings	Computer Organization and Architecture designing for performance	8th edition	Library of Congress Cataloguing-in Publication data on file	2010	ISBN-13: 978-0-13- 607373-4